School of Computer Science Engineering and Information Systems

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Department of Computer Applications

UBCA398J – Project – I

Review -1

Natural Scene Text Detection and Recognition

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ABSTRACT:

This project addresses the limitations of traditional Optical Character Recognition (OCR) systems when applied to natural scene images such as road signs, shop boards, and street nameplates. Conventional OCR engines like Tesseract are designed primarily for clean, document-style text and often fail in real-world environments with complex backgrounds, variable lighting, distortions, and arbitrary orientations, leading to low recognition accuracy and unreliable outputs. Existing approaches typically pass the entire image directly to the OCR engine without robust text localization, which causes misreads, missed text regions, and high sensitivity to noise and clutter. While these methods are simple and fast, their major drawbacks include poor performance on non-uniform scenes, inability to handle multi-oriented text effectively, and limited adaptability to real-world deployment scenarios.

To overcome these issues, the proposed system introduces a hybrid OCR pipeline that combines a deep learning–based text detector with a conventional OCR engine. The system first employs the EAST (Efficient and Accurate Scene Text Detector) model to precisely localize text regions in natural images, generating bounding boxes around candidate text areas. These cropped regions are then passed to Tesseract OCR for character recognition instead of sending the full image directly. This two-stage architecture significantly reduces background interference, improves robustness to noise, and enhances detection of multi-scale and multi-oriented text. The advantages of the proposed system include higher recognition accuracy on natural scenes, better handling of challenging conditions (shadows, clutter, skewed text), and improved practical usability for applications such as assistive tools for visually impaired users, smart navigation, and real-time translation of street-level text.

KEYWORDS: Computer Vision, Machine Learning, Artificial Intelligence

INTRODUCTION:

Text detection and recognition from natural scene images has become an important research problem in the fields of computer vision and artificial intelligence due to its wide range of real-world applications. Unlike scanned documents, natural scenes such as street signs, shop boards, billboards, and nameplates contain text embedded in complex backgrounds with variations in lighting, scale, orientation, font style, and noise. These

challenges make conventional Optical Character Recognition (OCR) systems ineffective when applied directly to real-world images. As smart cities, assistive technologies, and automated information systems continue to evolve, there is a growing demand for robust and accurate scene text recognition solutions. This project focuses on addressing these challenges by integrating deep learning–based text detection with traditional OCR techniques to improve the reliability and accuracy of text recognition in natural environments.

PROBLEM STATEMENT:

Traditional Optical Character Recognition (OCR) systems are primarily designed for clean, document-based images and perform poorly when applied to natural scene images containing complex backgrounds, varying illumination, distortions, and multi-oriented text. When the entire scene image is directly processed by conventional OCR engines, irrelevant background information, noise, and clutter lead to inaccurate text recognition, missed text regions, and unreliable outputs. Existing approaches lack robust text localization and struggle to handle real-world conditions such as shadows, skewed text, and non-uniform surfaces. Therefore, there is a need for an efficient and accurate scene text detection and recognition system that can precisely localize text regions and improve recognition performance in natural images for practical real-world applications.

OBJECTIVES:

The primary objective of this project is to develop an efficient and reliable system for detecting and recognizing text from natural scene images. The project aims to overcome the limitations of traditional OCR systems by integrating a deep learning–based text detection approach with a conventional OCR engine. Specifically, it focuses on accurately localizing text regions using the EAST text detector and improving recognition accuracy by processing only the detected text regions instead of the entire image. The system also aims to enhance robustness against real-world challenges such as complex backgrounds, varying lighting conditions, noise, and multi-oriented text, thereby making the solution suitable for practical applications including smart navigation systems, assistive technologies, and automated information extraction.

SCOPE OF THE PROJECT:

The scope of this project is limited to the detection and recognition of textual information present in natural scene images such as street signs, shop boards, billboards, and nameplates. The project focuses on implementing a hybrid OCR pipeline that combines the EAST deep learning–based text detection model with the Tesseract OCR engine to improve recognition accuracy in complex real-world environments. The system addresses challenges such as background clutter, varying illumination, noise, and multi-oriented text, while supporting commonly used scripts and fonts. The scope includes image preprocessing, text localization, and character recognition, but does not extend to handwritten text recognition, multilingual translation, or video-based text extraction. The proposed system is intended for offline image processing and research-level evaluation rather than large-scale commercial deployment.

PROPOSED SYSTEM:

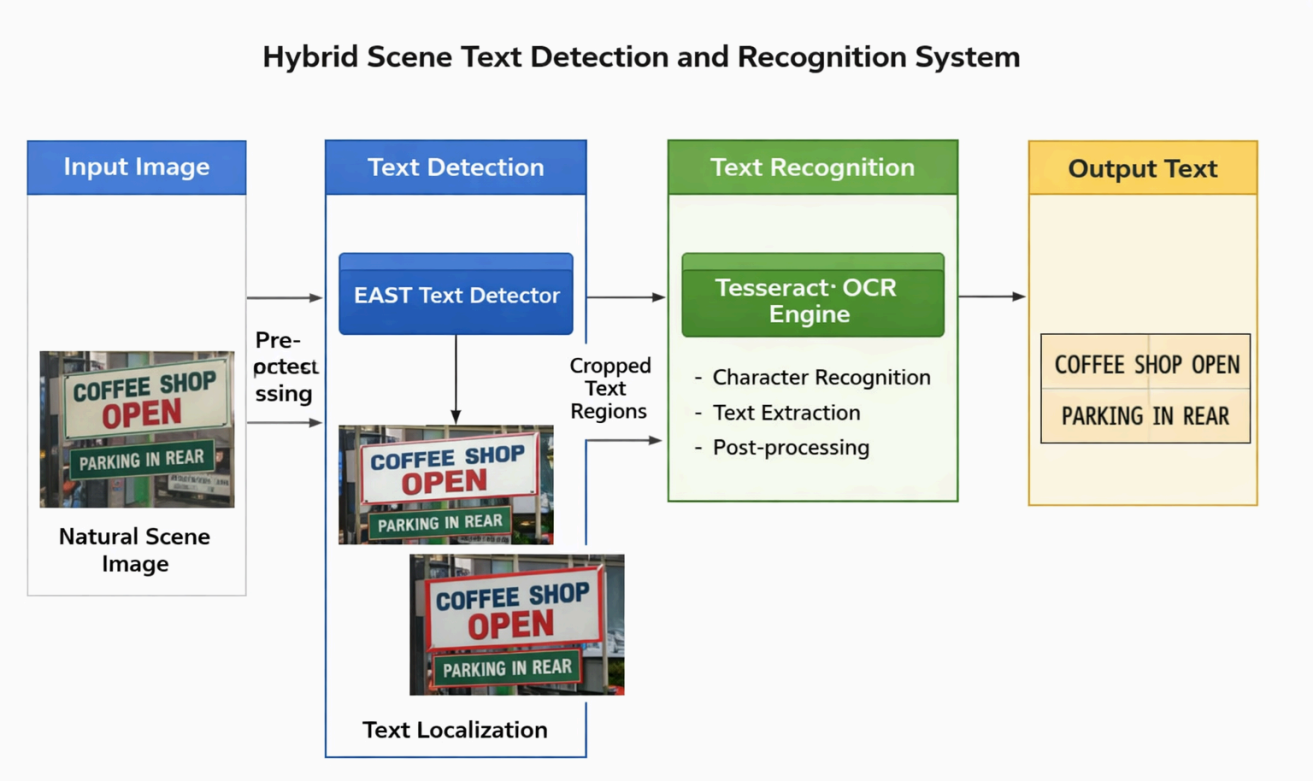
The proposed system introduces a hybrid scene text detection and recognition framework designed to improve the accuracy and reliability of text extraction from natural scene images. Instead of directly applying a conventional OCR engine to the entire image, the system first employs the EAST (Efficient and Accurate Scene Text Detector) deep learning model to accurately identify and localize text regions within the scene. The detected text areas are then cropped and passed to the Tesseract OCR engine for character recognition. This two-stage approach significantly reduces background interference, minimizes noise effects, and enhances the recognition of multi-scale and multi-oriented text. Additionally, image preprocessing techniques are applied to improve text clarity before recognition. The proposed system offers improved robustness under real-world conditions such as varying illumination, complex backgrounds, and distortions, making it suitable for practical applications like smart navigation, assistive systems, and automated information extraction.

LITERATURE SURVEY:

S.NO	TITLE	MERITS	DEMERITS
1	Scene Text Detection in Natural Images: A Review	Comprehensive overview of detection datasets, methods, and challenges in natural scenes.	Focuses on detection only, not recognition pipeline.
2	From Detection to Understanding: A Systematic Survey of Deep Learning for Scene Text Processing	Structured classification of detection and recognition models and benchmarks.	Limited practical hybrid system implementation.
3	Text Recognition in the Wild: A Survey	Thorough summary of scene text recognition challenges and methods.	Less emphasis on text localization techniques.
4	EAST: An Efficient and Accurate Scene Text Detector	Fast and accurate multi-oriented text detector, suitable for many benchmarks.	Only detects text regions; requires OCR integration.
5	TextOCR: Towards Large-Scale End-to-End Reasoning for Arbitrary-Shaped Scene Text	Provides a large annotated dataset improving OCR training and evaluation.	Focuses on dataset and annotation more than novel algorithm design.
6	PSENet: Shape Robust Text Detection	Effective at detecting irregular and curved text shapes.	Computationally heavy and slower inference.
7	Vision-Language Pre-Training for Boosting Scene Text Detectors	Introduces multimodal pre-training that improves detection robustness across models.	Pre-training increases computational and data requirements.
8	A Survey on Text Recognition from Natural Scene Images (IJERT)	Broad comparative study of text recognition methods for scene images.	Does not cover cutting-edge deep learning transformers.
9.	Scene Text Recognition with Self-Supervised Contrastive Predictive Coding	Shows advantages of self-supervised pre-training for recognition robustness.	Requires careful pre-training strategy and design.
10	Display-Semantic Transformer for Scene Text Recognition	Transformer based model with linguistic interaction improves STR accuracy.	Transformer models can be computationally intensive.
11	FOTS: Fast Oriented Text Spotting with Unified Network	End-to-end text spotting model combining detection & recognition.	Still struggles with heavily curved or arbitrary shaped text.
12	STN-OCR: A Single Neural Network for Text Detection & Recognition	Unified OCR model requiring no separate detection/recognition pipelines.	Performance is lower on complex natural scenes.
13	Deep TextSpotter: End-to-End Scene Text Localization & Recognition	Jointly trains detection & recognition improving overall accuracy.	Older architecture now superseded by transformer-based systems.

14	TextSnake: A Flexible Representation for Detecting Curved Text	Specialized for irregular curved text detection with good geometry modeling.	Complex geometry representation increases model complexity.
15	MORAN: A Model for Rectification & Recognitio of Text in the Wild	Improves recognition of distorted text via rectification before recognition.	Can struggle with heavy background clutter without strong detection upstream.

SYSTEM ARCHITECTURE:



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